Implementing Privacy-Preserving Techniques for Secure Healthcare Data Analytics

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Objectives

The main objective of this project is to develop a privacy-preserving healthcare data analytics framework that enables secure data sharing between multiple hospitals without exposing sensitive patient information. Healthcare datasets contain confidential data such as patient records, diagnosis details, and treatment history, which must be protected during data processing and analysis. This project aims to implement a two-level security mechanism using Denoising Autoencoder and Adaptive Transform Level Fusion encryption to protect the structural and statistical properties of healthcare datasets. The system allows hospitals to share encrypted data with a central server, where secure data fusion and machine learning analytics can be performed while maintaining data privacy.

Existing System

n traditional healthcare data analytics systems, datasets from different hospitals are collected and stored in a centralized server for analysis. These systems often rely on basic encryption techniques or simple data protection methods. However, during data sharing and analytics processes, sensitive patient information may still be exposed. Existing systems usually provide single-layer security, which is not sufficient to prevent statistical attacks or pattern-based data leakage. Additionally, these systems do not effectively hide the structural relationships within the datasets, making them vulnerable to privacy breaches. As a result, there is a need for a more advanced privacy-preserving framework that can secure healthcare data while still enabling effective analytics.

Proposed System

The proposed system introduces a two-level privacy-preserving architecture for secure healthcare data analytics across multiple hospitals. In the first level, each hospital processes its dataset using a Denoising Autoencoder, which removes noise and encodes the dataset into a latent representation, providing structural protection. The encoded data is then encrypted and transmitted securely to a central server. In the second level, encrypted datasets from different hospitals are combined using Adaptive Transform Level Fusion, which applies value-level transformation, position-level mixing, and dynamic key injection to enhance security. During data retrieval, the system performs decryption using reverse operations such as reverse bit rotation, reverse nonlinear mapping, reverse permutation, and autoencoder decoding. This dual-layer approach ensures high data privacy while enabling secure machine learning analytics.

System Requirements

Hardware Requirements

The system requires a computer with a minimum Intel i5 or equivalent processor, 8 GB RAM, and sufficient storage capacity such as 256 GB SSD to handle dataset processing and machine learning computations.

Software Requirements

The software environment includes Python programming language for implementing machine learning models and encryption algorithms. Development tools such as Jupyter Notebook or Visual Studio Code are used for coding and experimentation. Libraries such as TensorFlow, NumPy, Pandas, Scikit-learn, and Matplotlib are used for data processing, model development, and visualization.

Literature Survey

S.No	TITLE	AUTHOR(S)	INFERENCE	PUBLISHED IN
1.	Privacy Preservation In Artificial Intelligence - Enabled Healthcare Analytics	Shancang Li1* , Muddesar Iqbal2 , Ali Kashif Bashir3,5,6 and Xinheng Wang4	privacy-preserving methods improve secure data storage and controlled access in healthcare systems.	RESEARCH (Research Manuscript) Open Access
2.	Privacy - Preserving AI For Secure Data Sharing In Healthcare	Rajashekhar Reddy Kethireddy	encryption-based privacy techniques can securely enable healthcare data analysis without compromising patient confidentiality	Journal of Recent Trends in Computer Science and Engineering (JRTCSE)
3.	AI- Driven Secure Data Sharing: A Trustworthy And Privacy - Preserving Approach	Al Amin∗ , Kamrul Hasan∗ , Sharif Ullah† , Liang Hong∗ ∗Tennessee State University, TN, USA	encrypted data can be effectively used in advanced machine learning models while maintaining high privacy and accuracy.	University of Central Arkansas, Conway, AR, USA
4.	Privacy-Preserving Machine Learning For Healthcare: Open Challenges And Future Perspectives	Alejandro Guerra-Manzanares∗ , L. Julian Lechuga Lopez∗ , Michail Maniatakos and Farah E. Shamout	combining security mechanisms with machine learning enhances reliable and safe healthcare predictions.	ICLR 2023 Workshop on Trustworthy Machine Learning for Healthcare

Architecture Diagram:

Start

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Patient / Healthcare Data Collection

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Preprocessing of Healthcare Data

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AES Encryption of Sensitive Data

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Secure Storage of Encrypted Data

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Authorized User Authentication

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Decryption Using Secret Key

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Machine Learning Model (Logistic Regression)

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Disease Prediction / Data Analysis

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Secure Result Display to Authorized User

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End

Modules

The project consists of several modules that have been developed and implemented during the project development process. The Data Collection Module gathers healthcare datasets from multiple hospital sources for analysis. The Denoising Autoencoder Module processes the datasets to remove noise and generate encoded representations. The Encryption Module secures the encoded datasets before transmission. The Secure Data Transmission Module ensures that encrypted datasets from different hospitals are safely transferred to the central server. The Adaptive Transform Level Fusion Module combines encrypted datasets from multiple sources while maintaining privacy protection.

Work Plan & Timeline

Phase	Activities	Duration	Phase	Activities
Phase 1	Problem Definition & Literature Survey	2 weeks	Phase 1	To securely analyze healthcare data using machine learning while preventing privacy breaches and unauthorized access
Phase 2	Design & Implementation	4 weeks	Phase 2	Data masking,Anomaly detection ,Noise addition
Phase 3	Techniques Used	4 weeks	Phase 3	AES technique is used in this part and by that we are encrypting the datas
Phase 4	Model Training & Evaluation	1 weeks	Phase 4	Machine learning model like random forest technique is used

THANK YOU